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Climate discourse and economic downturns: The case of the United States, 2008–2013

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ABSTRACT

Global progress on climate change has been stalled since the 2008 financial collapse. I consider the case of the United States and how the crash and subsequent stagnation have reduced prospects for climate solutions at the national level. I focus on discourse and the economic and cultural framing of climate protection and the environment as a luxury good, unaffordable in recessionary times. As a solution to this paralyzing framework, I consider the emergence of "new economics," which employs an alternative framing at the local and regional level. New economics is fostering the creation of new forms of ownership in a range of enterprise structures that also contribute to sustainability and greenhouse gas reductions.
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1. Introduction

It is now seven years since the September 2007 run on the British bank Northern Rock. It is six years from the panic that gripped the markets and triggered a worldwide economic collapse and four years from the trough of that recession. While it is too early for a thorough accounting of the ways in which the financial crisis and subsequent downturn have affected the transition to sustainability in the global North, a number of effects are already apparent.

At the time the global financial panic hit, in the fall of 2008, the world's scientists and environmental policymakers were preparing for the climate summit in Copenhagen. In the months leading up to the

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summit, scientists produced a series of reports on the state of the planet. They spoke with a single voice: things are worse than we anticipated at the time of the last IPCC report in 2007. The international community went to Copenhagen with unprecedented optimism and determination that rationality, compassion and perhaps even the survival instinct would prevail. The watchword of the week was Hope—who can forget the renaming of the city to "Hopenhagen?" Today, it feels naïve to have believed in the possibility of a global climate solution at the very moment the world economy was collapsing around us.

After February, the global economy got worse and so too did the situation with climate. The global emissions trajectory has been relentless. Emissions are nearly 60% higher than they were in 1990, the benchmark year enshrined in the Kyoto Protocol, and rising emissions mean that the 2° target is unlikely to be met (Global Carbon Project, 2012). In 2012 emissions grew 2.5%, and in 2013 the atmospheric concentration of carbon dioxide reached 400 parts per million, considerably above the 350 ppm that many scientists say is a safe limit (Earth System Research Laboratory, 2013; Hansen et al., 2008). Modest emissions declines in wealthy countries have been insufficient to counter-balance the large increases in the global South, especially from China and India. And some of what is easing emissions in the United States, such as the shift from coal to hydraulic fracturing for natural gas, is accompanied by gas flaring and methane release that may turn out to be worse than the practices they are replacing (Howarth et al., 2011).

Meanwhile, the effects of climate disruption have become increasingly visible. In 2012, Arctic ice hit a new low, putting the predicted date for an ice-free Arctic within a decade, rather than the 100-year timeline originally expected by scientists (Gillis, 2012). In March of 2012, 15,000 temperature records were broken in the United States, as the average temperature was 8.6 °F above the 20th Century March average (US Department of Commerce, 2012) and the continental U.S. suffered its worst drought in 50 years (Masters, 2012; see also NOAA National Climatic Data Center 2012). In October, Hurricane Sandy, possibly the worst storm ever to hit the Northeast United States, devastated the region (Wikipedia, 2013a). The year 2012 now ranks as the hottest in US history, by an astonishing full degree Fahrenheit (Gillis, 2013).

During this period, the US and global economies have struggled, and failed to regain their earlier growth trajectories. In the US, the original reduction in national output was severe, in comparison to other OECD countries. GDP fell by 12% in the six months after the financial panic (Council of Economic Advisors, 2013) and while growth in GDP resumed by 2010, annual growth rates have been below 3% (Council of Economic Advisers, 2014, Table B-1, p. 367). While GDP has grown, the fraction of the civilian population that is employed plummeted from 63.0% in 2007 to 58.5% in 2010, and has barely risen since then. (Council of Economic Advisers, 2014 Table B-11, p. 379) Unemployment and under-employment, which rose to almost 27 million at the height of the downturn, remained at nearly 22 million in 2013 (Economic Policy Institute, 2013). Continuing economic weakness has contributed to rising poverty: the number of Americans relying on government assistance to meet their basic food needs continues to increase, rising to 46.6 Million in 2012, up from only about 16 million in 2000 (Wikipedia, 2013b).

Using the case of the United States, I will argue that the downturn has severely crippled attempts to move toward sustainability, particularly in the area of climate. One reason is the framing of environmental protection as a luxury, affordable only in good times. "Environment as luxury" is a longstanding discourse that has been difficult to dislodge; it exerts an especially paralytic effect during periods of economic distress. In this paper I focus mostly on issues of framing, or discourse, rather than the flows of investment into renewable energy or other green initiatives. An extensive literature in sociology finds that framing is central to the success of social movements and their attempts to make social change (Gamson et al., 1982; Snow et al., 1986; Snow and Benford, 1988; see Benford and Snow, 2000 for a review. See also Nisbet (2009), who has used frame analysis to discuss climate change.) I will argue that climate change has been "framed" poorly and in a way that makes it vulnerable to counter-framings, such as the need for jobs or economic growth.

My argument is consistent with papers from the special issue of this journal on the impacts of the financial crisis, and in particular with the detailed analysis of Geels (2013). Discussing Europe and to a lesser extent the United States, Geels argues that after the early stimulus funds, which included large funding streams for renewable energy, the "window" of opportunity began to close in 2010–11, and

since then the financial crisis has continued to impede the transition to sustainability. Furthermore, my analysis is along the lines of Antal and van den Bergh (2013) in that it is oriented to macroeconomic dynamics. While this paper takes up the case of the United States, it may also be a cautionary tale for those parts of Europe that are suffering from stagnation and high levels of unemployment. It is also consistent with the arguments of Scruggs and Benegal (2012).

While there is considerable reason to be pessimistic about both the state of climate politics and discourse in the United States, and the nation's willingness to play a leadership role in global climate policy, there are some hopeful developments. One is the emergence, albeit at the margins of mainstream discourse, of a new approach to economic development that employs an alternative to the dominant tradeoff framing, which has the possibility of re-aligning the relationship between environment and economy. In the US, it is being called "new economics," and it is emerging from devastated inner-cities and stagnant regions as a solution that both revitalizes local economies and addresses climate and other eco-system degradation. New economics initiatives include green worker cooperatives focusing on democratizing ownership; renewable energy consumer cooperatives; local, organic food systems; peer-to-peer sharing platforms that reduce consumer demand for new goods; and a variety of other economic innovations with the multiple aims of reducing carbon footprints, reducing scale and democratizing the economy, and redressing inequality. From the perspective of this paper, the defining feature of these initiatives is that they reject tradeoff thinking and consider reduced environmental impact as a component of successful economic activity, rather than an alternative to it.

2. Climate politics in a time of economic weakness

2.1. National climate policy in the wake of the financial crash

The persistence of economic weakness has had significant effects on the discourse about climate and the environment more generally. The early period of the downturn—2009–2010—did not yield the progress on climate that one would have expected on the basis of the importance Obama placed on this issue during his Presidential campaign. For example, in his 2008 nomination acceptance speech, Obama said that if we worked hard he was positive that "this was the moment when the rise of the oceans began to slow and our planet began to heal. . ." (Obama, 2008). While there was about $50 billion of "green" spending in the original stimulus bill (out of a total of nearly $800 Billion), political appetites to address climate and environment disappeared rapidly (Schor, 2010). No major additional stimulus money for a clean energy transition was forthcoming, and by 2012 Republicans were on the offensive against government support for renewable energy companies. (The outsized attention given to the Solyndra bankruptcy in the political arena was indicative of the changed political landscape.)

In 2009 the House of Representatives passed the first climate legislation in the Congress, a cap and trade bill referred to as Waxman-Markey. However, it proved irrelevant when the Senate decided not to consider a companion bill. Energy interests and their political allies spent an estimated $500 million lobbying against the bill from January 2009 to June 2010 (Weiss et al., 2010). Their public campaign argued that it was a job killer that would reduce standards of living and cause economic distress, a message that was far more powerful when unemployment was at double-digits and GDP was plummeting. Furthermore, the large giveaways of free permits to polluters in the bill meant that consumers were likely to bear the costs of higher energy, in contrast to alternative financing schemes in which higher energy costs are rebated directly to the public (the cap-and-dividend option) or into public revenues. Of course, one could say that progress on climate has failed primarily because of the power of the oil, coal and gas industries and their allies, and I would agree with that view.

But it is also true that the economic collapse provided the fossil fuel industry with an opportunity that it did not have previously. The economic situation was a game-changer that climate deniers and the fossil fuel industry used to engineer an aggressive U-turn on the state of play of climate legislation. Just a short while earlier, Republican opposition to legislation was not inevitable. Prominent Republicans had shifted to agreeing that the Federal government needed to act on climate. Indeed, the

cap and trade bill that came out of the House was originally a Republican bill and backers expected bi-partisan support (Skocpol, 2013). It seems that at least a significant number of Republicans had accepted the idea that climate legislation was going to be enacted. The other effect of the financial collapse was that it eroded support for the complex trading mechanism included in the bill—the idea of a major government program that would set up a new market to enhance profit opportunities for Wall Street was a very hard sell at that time, given the public fury at financial institutions. Thus, it seems clear that although the politics of climate legislation are complex, the "Great Recession" was a major factor in the legislative failure, although one that analyses by political scientists have mostly failed to appreciate (Skocpol, 2013; Weiss, 2010).

2.2. The 2010 midterm elections and their aftermath

Things changed dramatically with the 2010 midterm elections and the Republican takeover of the House of Representatives. The election reflected both normal midterm losses for the Presidential party as well as public dissatisfaction with economic conditions. The House of Representatives voted in at that time has been the most anti-environmental in history. Between January 2011 and June 2012, nearly one out of every five votes in the House was to undermine environmental protections—315 votes out of a total of 1535. The House took 37 votes against action on climate. They took 115 votes to de-fund the EPA. They attempted to dramatically weaken clean air and water legislation, taking 77 votes against the Clean Air Act, and delays in new emissions and air pollution regulations. They even voted down an amendment that would require Federal regulators to seek advice from at least one source other than the American Petroleum Industry (Democrats Committee on Energy and Commerce, 2012; US Congress Votes Database, 2013). The House took 109 votes to benefit oil and gas companies and is more indebted to major oil, gas, coal and other extractive industries and their political allies than any congress in history (Weiss and Weidman, 2012). They bullied the President into delaying clean-air regulations and jeopardized funding for clean energy—a policy with broad popular support.

The extent to which the recession undermined attempts to address climate change can also be seen in the conduct of the 2012 Presidential election. Mitt Romney, who as Governor was active in putting together a regional climate initiative, not only turned into a climate denier, but in his convention speech mocked Barack Obama's mere mention of rising sea levels. For his part, Obama continued to avoid political risk by espousing his disastrous so-called "all of the above" energy policy, i.e., promotion of all energy sources, including renewables and fossil fuels. Climate change was apparently such a fearsome political issue that candidates maintained near-total silence throughout the campaign, treating it as "that which shall not be named," (borrowing from a phrase from the hugely popular Harry Potter series that was used to refer to the awesomely powerful and evil Voldemort). That changed a bit when Hurricane Sandy hit, and Mayor Michael Bloomberg of New York City endorsed Obama specifically because he was better on climate. Silence on climate was also the strategy of the mainstream media. Throughout 2012 less than one hour of coverage was given to climate issues by all the major nightly news shows (excluding stories on the damage and human suffering wrought by super storms), and of that hour less than 10 minutes addressed scientific findings (Fitzsimmons, 2013).

The effects of economic stagnation on climate discourse and policy noted above have been in the political arena. There are of course many economic impacts, too detailed and complex for me to explore in this paper. However a few may be worth mentioning. Declines in GDP are a reliable way of cutting GHG emissions, and this downturn proved no different (York, 2008; Knight and Schor, 2014). U.S. emissions have fallen significantly and in 2012 they were an estimated 17% below the business-as-usual baseline scenario. The Council of Economic Advisors estimates that 52% of the reduction is attributable to the recession, 40% is due to fuel switching (out of coal, into natural gas and wind), and the remaining 8% is from increased efficiency (Council of Economic Advisors, 2013, pp. 194–96). Second, the downturn also reduced energy prices—this has a complex impact over time, affecting rates of exploration and substitution. Third, the rate of investment in renewable energy sources slowed considerably and initially declined with the downturn. U.S. Investment in clean energy plummeted in 2009. By 2010 it had recovered, but again declined nearly 40% in 2012 (National Science Foundation, 2012; The Pew Charitable Trusts, 2012).

3. Framing climate and the environment

3.1. Economists' framing of the economy-environment relation

Why did the economic downturn and high unemployment have such a strong impact on willingness to pursue environmental and climate goals? After all, at least since the publication of the Stern Report in 2006 even a discipline as conservative as economics agrees with the view that not addressing emissions is more costly than mitigation (Schor, 2010). There is not a credible economic argument against a forceful response on climate, and even leading conservative economists favor carbon taxes or similar policies (Mankiw, 2013). I believe that part of the answer is the larger framing of the environment in both mainstream political discourse and economic theory. On the one hand economists treat environmental externalities as market failures, which means that interventions to internalize them should improve welfare. Economists should therefore welcome environmental protections and climate mitigation. However, they also often employ a contradictory discourse that suggests that protecting the environment requires reducing welfare because it sacrifices other goals such as income, jobs, or profits (Schor, 2010).

For the purposes of evaluating the impact of economists' framings on public discourse and policy, I consider textbook presentations and the metaphors used in popular discourse, as my intention is to discuss the ways in which economists communicate with the public about these issues. Furthermore, I discuss the "environment" rather than "climate" because most economic treatments, and especially in textbooks, are framed with respect to the environment more generally. Within economic theory, the main metaphor that is employed for thinking about the relationship between environmental protection and well-being is that the environment is a normal consumption good. One buys more "environment" when one has higher income. (For a discussion of this framing, see Schor, 2010, ch 3.)

A common formulation of this kind of thinking is the Environmental Kuznets Curve, a hypothesized relationship aiming to show that as countries begin to grow they degrade the environment as a way of creating jobs, income and prosperity (Grossman and Krueger, 1995). Over time, as income increases, public pressure for more protection is registered through the democratic political process by voters who now value the environment more highly. This relationship is analogous to the original Kuznets curve, which posited the same mechanism between growth and income inequality. Countries tolerate more inequality in the initial stages of development and then the public demands more "fairness" as the society gets wealthier. The Environmental Kuznets Curve was originally based on the cases of single toxic gases such as sulphur dioxide that are regulated nationally, but is typically not supported for global and large-scale environmental problems such as climate change, ecological overshoot or biodiversity. (See Caviglia-Harris et al., 2009 or the discussion in Schor, 2010.) But my purpose here is not to debate the veracity of the EKC—it is to note the importance of this type of thinking in the economic discourse and the public mind.

A related metaphor is the production possibility frontier, the standard textbook presentation of the environment. It is a similar story to the Environmental Kuznets Curve, but for one point in time (rather than a story about development over time). The production possibility frontier says that countries that want more environmental protection have to give up consumer goods. In this way the environment is theorized as a good itself. Opting for more environmental protection yields less in the way of income and goods.

Of course, not all economists have accepted these framings. In the United States, Goodstein (1999) has provided a comprehensive empirical critique of the tradeoff formulation. Martinez-Alier (1995) has argued for the "environmentalism of the poor," which is a clear contrast to the idea of the environment as a luxury good. Similarly, Boyce's work on "natural assets" combines standard economic concerns with distribution (Boyce and Shelley, 2003; Boyce, 2013). The field of ecological economics itself represents a fundamental challenge to the idea of the environment as a consumer good, given that it treats the environment as a resource not only for production, but for life itself. Outside of economics, a variety of other framings, for example, environmental justice and climate justice, focus on the distribution of costs and benefits of environmental impacts and policies. However, within the discipline of economics itself, the environment/economics tradeoff remains the dominant approach, and alternative framings are relatively marginal.

3.2. Tradeoff framings in the public discourse

Although it may originate within economics, tradeoff framing is also common in the public arena, particularly since 2008. For example, soon after winning the 2012 election, President Obama provided a stark reminder of the grip of trade-off thinking. Asked by a reporter whether and how he intends to address climate change in his second administration, the President replied: "Understandably, I think the American people right now have been so focused, and will continue to be focused on our economy and jobs and growth, that if the message is somehow we're going to ignore jobs and growth simply to address climate change, I don't think anybody is going to go for that. I won't go for that" (Obama, 2012). This framing of environment versus jobs is extremely powerful and it has long been the dominant frame in the US political discourse, whatever the reality. Obama did go on to suggest there may be a way forward within the current paradigm: "If, on the other hand, we can shape an agenda that says we can create jobs, advance growth, and make a serious dent in climate change and be an international leader, I think that's something that the American people would support."[1]

The increasingly visible conflict over the Keystone XL pipeline is another example of how framing climate policy in tradeoff terms has undermined public support for climate protection. The Keystone XL pipeline would transport highly polluting tar sands oil, or bitumen, from Alberta, Canada through to the Gulf of Mexico. Canadian oil interests are desperate for the pipeline, as exit routes in their own country have been blocked by popular opposition. US economic interests in the pipeline are quite limited. Early estimates were as low as 6000 new jobs (Wald, 2013). While pipeline supporters have tried to produce additional studies to increase the job-impacts, they remain quite limited. And the bitumen is destined for other countries. But among those who have heard of it, 63% of the US population supports the pipeline (Leiserowitz et al., 2013b). In an environment of job scarcity, people are loathe to foreclose any opportunities for employment creation.

3.3. Tradeoff tropes among the public

Not only economists and politicians, but also ordinary citizens employ a tradeoff framing when discussing climate and the environment. In a Boston-based interview study of public attitudes toward climate change and ecological consumption conducted with my student, Laidley (2013), the trope of economic well-being versus the environment was commonly raised by informants. Concern for climate and environment was discussed as something that wealthier or professional people could afford. Respondents repeatedly articulated the opinion that if you are struggling to put food on the table you can't worry about global warming.[2] Most of the informants in this study took the view that concern for the environment is associated with higher levels of material security and education. In this sense, they subscribe to Abraham Maslow's psychological version of modernization theory (Maslow, 1943). While modernization theory is controversial for a number of reasons, including its roots in Orientalist and racist discourses (Said, 2003 [1978]), popular versions of this view remain deeply imprinted on the minds of the public. Maslow's hierarchy of needs theory holds that a person has to be materially secure, with good employment, to have moral and spiritual concerns. The Maslovian view that poor people don't and can't care about the environment is a deeply held myth in the popular imagination. The realities that high income individuals, households and nations have much larger eco- and carbon footprints, or that the poor are frequently the fiercest protectors of forests, water resources, and eco-systems, are discounted in the popular imagination. There is widespread resonance of the idea that concern for the environment is a luxury, affordable only by the wealthy.

[1] After this paper was written, the President gave a strong speech on climate in which he repudiated the tradeoff message. Time will tell whether he follows that rhetoric with action.

[2] Some, like Tim, an unemployed 51 year old former city employee, thought that concern for the environment by the privileged while the economy is doing so poorly leads to a backlash and a tendency to portray environmentalists as hypocrites: "You know, that people are struggling just to get by and you've got people, with a lot of money, pushing organics, food, and stuff. I think it's hypocritical" (Laidley, 2013). Jessica, a relatively low income student says green consumption and environmentalism "Seems to be more of an upper-class thing. . .Probably because they have more money to spend on the. . .the changes that you need to make in your life to buy those expensive products. And like buy different cars, (they) probably can afford it more. Maybe that's why they believe (in) it (climate change)" (Laidley, 2013).

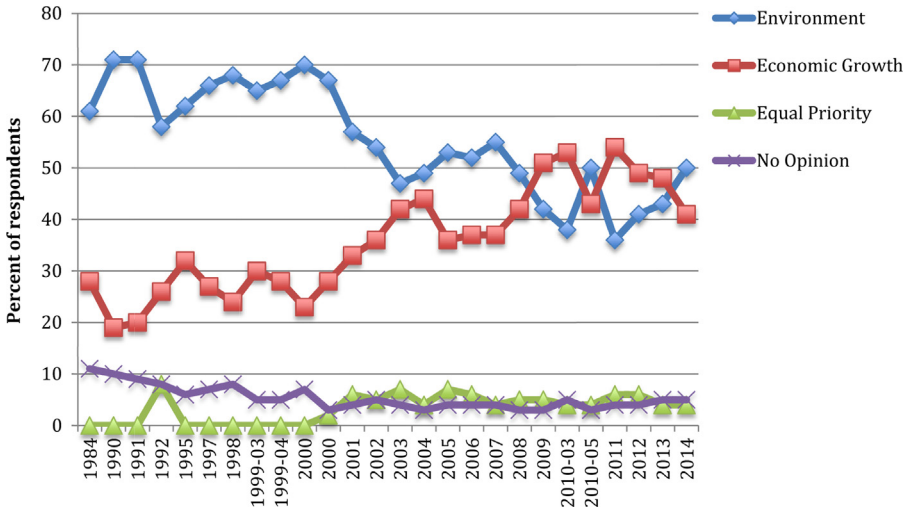

Fig. 1. Public opinion on environment versus economic growth. *Note:* Responding to query of whether environment, even at risk of economic growth, should be prioritized or whether economy, even at risk of environment should be prioritized. Data from Gallup: http://www.gallup.com/file/poll/168038/Environment_vs_Economic_Growth_140320.pdf.

Research by my colleague Holt (2014) also supports this interpretation. After the crash, Holt conducted in-depth interviews with middle and lower middle class households, exploring their economic experiences and attitudes toward sustainability. He found deep economic distress and rapid downward mobility, pessimism, and a failure of imagination for alternatives. He also found that few of his informants give any thought to sustainability and see it as something that is not relevant to their lives. Lifestyle versions of sustainability have no credibility, and are seen as relevant only to the privileged.

3.4. Trends in public opinion on climate and environment

We can also look to the timing of changes in public opinion as evidence of the impact of the recession and the power of tradeoff thinking on Americans' views on growth and the environment. Scruggs and Benegal (2012) have studied the impact of the Great Recession on attitudes toward climate change. In both the United States and Europe, the recession had a strong negative impact not only on concern for climate change, but also on people's basic beliefs about whether it is occurring. Their model controls for a variety of other factors (demographics, party affiliation, climate denier activity, etc.) and finds that with temperature fluctuations, the unemployment rate has a large and significant impact on polling results. Here I will merely present trends in the public opinion data, which I interpret similarly to Scruggs and Benegal (2012).

For decades, the Gallup polling organization has been asking Americans which they place a higher priority on—economic growth or protection of the environment. (There is no comparable series for climate so I present evidence on attitudes toward "the environment.") Fig. 1 presents this series, and it shows strong support for the view that current economic conditions affect environmental attitudes. Since the question was first asked, the environment had always been prioritized over "economic growth." However support for the environment began to erode sharply in 2007, as the sub-prime mortgage crisis rippled through the economy and various regions started to experience a housing price collapse and GDP decline. In 2009, at the depths of the recession, and for the first time ever, a majority opted for economic growth over protecting the environment. As the economy began to recover, Americans increasingly began to prioritize the environment again. In 2014, after five years of recovery, the environment was again listed as more important than economic growth, by 9 points (Swift, 2014).

A similar story can be found in the timing of attitudes on global warming and climate change. While the Gallup numbers are about priorities and values, climate change polling also includes questions about basic beliefs. Do you believe that global warming is occurring? Is it man-made? Do scientists agree about the existence of global warming? The Great Recession coincided with a large shift in Americans' views of whether global warming was actually occurring. The Yale Center on Climate Change Communication provides extensive data on beliefs about climate change and preferences for action. Between November 2008 (just as the crisis came into full view) and January 2010, belief that "global warming is happening" fell an astonishing 14 points—from 71% to 57%. By September 2012 most of that drop had been restored, although in the last year the number has again dropped (to 63%). In 2013, belief in global warming was still a full 8% below 2008 levels (Leiserowitz et al., 2013a, Appendix 1).

Similarly, attitudes about how the government should respond to global warming also shifted with economic conditions. The fraction of the population that thinks global warming should be a high or very high priority for the President and Congress fell from 54% in November 2008 to 42% in April 2013 (Leiserowitz et al., 2013b, Appendix 1).

3.5. Concluding thoughts on framing

The experience of the last six years suggests that how climate and the environment are framed is important (Nisbet, 2009). The dominant frame has been that environment and climate are consumption goods, affordable only in prosperous times. When hard times arrive, climate and environmental protection are cast aside in favor of policies that are aimed at accelerating GDP growth and creating jobs. Alternative framings, in which emissions reductions and environmental investment are seen as means of job creation and welfare improvement, failed to gain sufficient momentum to avoid regression on climate attitudes and policies in the aftermath of the crash. To advance the cause of climate and eco-system protection, we need a broad, compelling narrative in which prosperity and well-being are intimately connected to maintaining the health of the climate and natural environments, locally, regionally and globally. I now turn to discuss attempts to create such narratives.

4. New discourses of climate and economy

4.1. Mainstream discourses of green jobs and green growth

I have argued that the dominant economic discourse both before and after the crash has been that the climate (or environment) and the economy must be traded off against each other. In recent years, there have been attempts to put forward alternative framings. In the United States, the most prominent has been the concept of "green jobs." (Jones, 2009; Jones and Conrad, 2012; Pollin, 2012a,b) The core of the green jobs approach is that both the transition to clean energy and improvements in energy efficiency create jobs and incomes. This is because of direct government expenditures and also because renewables are more labor-intensive than fossil fuels. This idea has been influential in particular regions, for example in the Midwest, where wind energy has been seen as a solution to collapsing farm incomes and manufacturing capacity. However, while Federal green jobs funding was contained in the original stimulus bill, such funding was never repeated. An illustration of the short life of this approach is that Van Jones, the White House advisor on green jobs, was attacked by Republicans and pushed out in 2009.

A related framing is the "green growth" paradigm (Fay et al., 2012; Friedman, 2008). This approach argues that the shift to renewable energy and a more general orientation to ecological sustainability will be the basis of renewed profitability and growth. Green technology will lead a general investment boom that will accelerate economic growth and competitiveness. The green jobs and green growth approaches both rely on a traditional approach, namely the use of government expenditures to enhance market processes.

In the United States, the green jobs and green growth rhetoric has been less successful than one might have expected, given its high-profile advocates and the fact that it is squarely within the mainstream of historical policy parameters. Why this is so deserves a study in itself. One reason may be

the failure to rate climate very highly as a problem; therefore solutions to it are not so likely to be taken seriously. It is also possible that the difficulty of reconciling carbon emissions targets with output growth is recognized as the daunting problem that it is. Few people are aware of evidence that shows a lack of de-coupling between carbon emissions and GDP (Jorgenson and Clark, 2012; Knight and Schor, 2014). However, there may be widespread intuition that achieving the radical emissions reductions that are necessary for climate protection is incompatible with continued growth in GDP (Anderson and Bows, 2011; Anderson, 2012).

Another reason may be that looking to renewable energy as a locomotive of prosperity is a largely technological solution that fails to address the larger systemic crisis of the US economy. To make this point I return to the argument I put forward in my 2010 book *Plenitude* (Schor, 2010). There I argued that the United States and European economies were facing a difficult period. The prediction in the book was that GDP growth would not quickly rebound to pre-crash levels and might never regain the 3–4% that characterized earlier decades. Second, emerging ecological scarcities have begun to permanently affect prices, as food, fuel and other commodities become more expensive. I argued that in the United States, unemployment would remain high, real wages and incomes for most people would stagnate or decline, and the prices of food and energy would escalate. All of those expectations have been borne out in the four years since the book was published. In the world of conventional economic policymaking, growth is the solution to unemployment, low wages and other economic problems. However, when growth fails to appear, there is no alternative policy to turn to, as neo-liberalism has ruled out most other options.

Policymakers and politicians have not yet faced up to this reality, however, it may be that among the public there is widespread understanding that the economic system is in systemic crisis. I have been unable to find public polling on this question. However, if this is the case, then proposed solutions that do not address underlying structural problems such as financialization, extreme concentrations of wealth and political power, technologically-induced unemployment and ecological overshoot may not be seen as credible.

4.2. The "new economics" movement: an emergent re-framing

In the last few years, a third approach has begun to emerge. Like the green jobs and green growth paradigms it is fundamentally reframing the relation between climate and economy. It has been called the "new economics movement." At the moment it remains marginal; indeed, it has not yet risen to the surface in the national discourse. However, it is gaining attention and adherents. I discuss it in more detail than the green jobs and green growth approaches because it is less well known than those.

The term "new economics" has been in use in the United Kingdom for quite some time, at least since the founding of nef, the new economics foundation, in 1986 (see also Seyfang, 2010). In the United States, it began to be used just after the financial crash. Activity began in the non-profit sector, where a number of national organizations using this language were formed. In 2009, the New Economy Network began and enlisted between 200 and 250 individuals and organizations to work for a new economy. (Alperovitz, 2011b). Not long after that the Schumacher Institute created a collaborative effort with the UK's new economics foundation (nef), which was called the New Economics Institute (since renamed the New Economics Coalition). In 2010, the Institute for Policy Studies convened the New Economy Working Group. In 2011, in the wake of the Occupy Movement, a group of undergraduates at Harvard walked out of the Introductory Economic class protesting its narrow, ideological slant (http://harvardpolitics.com/harvard/an-open-letter-to-greg-mankiw/) and a few months later convened a conference on "new economics." The claim that a "new economics movement" was developing was made in print by Gar Alperovitz in 2011 (2011b; see also 2012) and by Broad and Cavanagh in 2012, the year that the New Economics Institute held a major founding conference. Since that time there have been numerous conferences at colleges and universities around the country, sponsored by the New Economics Coalition. To date, scholars have paid relatively little attention to new economics.[3] There is not yet agreement among scholars as to what new economics consists of. However, on the

[3] In 2012 and 2013 I convened the "Summer Institute in New Economics" which is a week-long program for PhD candidates who are interested in new economics.

basis of my research and participation in the new economics movement I believe there are at least four principles that are common to all new economy initiatives, and a fifth which characterizes some.

The first is a radical rejection of the existing economic paradigm and approach to well-being. New economics is not focused on GDP growth as the mechanism for creating jobs or livelihoods. It is bottom-up, entrepreneurial, and grass-roots. It does not rely primarily on correcting the excesses of the market by ex-post tax and transfer policies. In that sense it is not fundamentally Social Democratic or Keynesian. It advocates whole system change on the grounds that global capitalism is an anti-human, unsustainable and dysfunctional system. Unlike the dominant climate discourse it is not primarily technological in nature—its focus is on deep economic and social transformation. Its view is that the economic system must be permanently and structurally transformed. At its core it references jobs, livelihoods and enterprise, along with other elements of well-being such as stable communities, healthy food systems, vibrant neighborhoods. It locates fairness at its center, arguing for more equitable distributions of wealth and power. It includes climate and eco-systems, but as part of an economic alternative, rather than just an ecological shift.

A key element of its rejection of the existing economic system has to do with a new approach to property. Social Democratic and Keynesian approaches typically focus on the relation between the market and the state: they use the state to improve market outcomes, and choose not to address the existing distribution of productive assets and financial wealth. New economics aims to fundamentally change the operation of the market, in large part through changing basic structures of asset ownership and the distribution of wealth and property. It takes the view that concentrations of economic power are detrimental because they allow small minorities to monopolize flows of value. Furthermore, highly concentrated economic power results in concentrated political power and subverts democracy (Alperovitz, 2011a). The benefits of more equal distributions of economic and political power will be more jobs, stable communities, and more democracy.

Therefore new economics focuses on a move away from corporate forms to a variety of ownership structures, such as worker, consumer, civic and housing cooperatives; land trusts; Community-Development Corporations (CDCs) and other novel enterprise forms that lead to a broader distribution of wealth. Legislation enabling a number of these forms, such as cooperatives, land trusts, and CDCs has long been in place. New legislation is also creating some newer forms. For example, beginning in 2010, states began passing laws authorizing Benefit-Corps, which are for-profit entities that are permitted to take social and environmental benefits into account in decision-making. There is also a strand of new economics devoted to alternative forms of financing such as public banks, community financing, crowd-sourced financing, impact investing and Slow Money. The larger strategy is to create institutions with alternative ownership structures, as well as to expand existing institutions that already have significant public or community ownership. As Alperovitz (2011a) details, a significant fraction of the US economy already consists of these kinds of enterprises, such as cooperatives, public utilities, municipally owned businesses, and credit unions, however the potential for public benefit from these enterprises is often not realized. In addition, proponents are attempting to expand this sector in towns and de-industrialized cities where there is considerable willingness to experiment with new forms of ownership and economic development because there is no opportunity for inflows of private investment. More broadly, new economics attempts to re-make the market so that it is characterized by fewer concentrations of power that lead to the exploitation of labor and the natural environment. This is often referred to as the "democratization" of wealth, i.e., the achievement of a more equal distribution.

Second, New Economics does not typically look for large-scale solutions, as its focus tends to be local and regional. New economists are not against large-scale policy interventions such as a global climate deal or a national carbon tax, and some are active in seeking such policies. However, much of the innovation is directed at the municipal and regional level where people believe they can be most effective during a period in which national action is not forthcoming because the state has been captured by corporations and the wealthy.

Third, new economics puts sustainability at its core. Initiatives have been in areas such as renewable energy, organic and local food system change, green manufacturing technologies, and the promotion of low carbon local economies. Indeed, one might characterize this movement as an alliance between

people concerned about inequality and social justice with those mainly concerned about climate and environment.

On the question of GDP growth and the debates about de-coupling, green growth and limits to growth, new economics has been mainly silent. In part this is because it has concentrated its efforts at the local and regional scale. There is a strong current of Schumacherian thinking in the new economics movement (Schumacher, 1973). However, it is also the case that, so far, new economics lacks a well-developed macroeconomic articulation. Some of its academic proponents are ecological economists with a strong concern for the scale of economic activity, and are therefore advocates of steady-state or de-growth. But others are either agnostic about growth or fail to address macroeconomic questions.

A final principle is one that only some parts of the new economics movement embrace. It is what I called being "technologically forward." What I mean by this is that digital technologies are being used and envisioned to create profound transformation of economic relations. One obvious area is scale: technological innovation has made possible a shrinkage of scale that was unthinkable in the 20th century (Schor, 2010). Small networked units are now a highly efficient option for the future. A second area in which technology is remaking economic and social relations is what has been called the sharing economy, for example peer-to-peer sites such as car sharing, lodging sharing, and time banking (Schor and Fitzmaurice, 2014). It also goes by the terms collaborative consumption and peer production. Sharing platforms bring a lifestyle dimension to new economics that is not present in a number of its core activities, especially those I noted above (e.g., new ownership structures and finance, reduction of scale, and ecological commitments). Sharing exchanges are thought to be carbon reducing, although that is not their primary intention, nor does it tend to be participants' primary motivation (Dubois et al., 2014).

4.3. New economics in practice

New economics is mostly a practitioner movement, rather than an intellectual enterprise. Its projects are expanding around the country. The most well-known are in the food sector, where alternatives to industrial agriculture are exploding: in addition to organic and local farming using sustainable methods, new structures at the local level such as Community Supported Agriculture are expanding rapidly. Restaurants are committing to local food procurement. Regional food hubs and networks are forming. But there is also considerable change outside of agriculture. In the energy sector, a variety of new enterprise forms in renewable energy have emerged. These include community solar installations, which are putting solar cells on community buildings, and are owned by small investors. Crowd funding of solar energy has also grown rapidly. Other examples are innovative consumer cooperatives such as New England's Coop Power, which is a social-justice oriented incubator of member-owned innovative businesses across the renewable energy sector. In Western Massachusetts, the group that invented the CSA is now working on expanding the concept to Industry. And worker cooperatives are being formed in solar energy installation and retrofit, biomass, and other fuels.

According to Gar Alperovitz, a leading new economy scholar and innovator, the new economy is much larger than is generally recognized. "One hundred thirty million Americans, in fact, already belong to one or another form of cooperative—and especially the most widely known form: the credit union. Similarly, there are some 2000 municipally owned utilities, a number of which are ecological leaders. (Twenty five percent of American electricity is provided by co-ops and public utilities.) Upwards of 10 million Americans now also work at some 11,000 employee-owned firms (ESOP companies)." (Alperovitz, 2012)

One of the most well-known and successful projects is the Cleveland Model, a novel arrangement in which a series of green worker cooperatives are being set up with support from large "anchor" institutions—the city's educational and medical institutions—and the local civic philanthropy and the city government (Alperovitz et al., 2010). The Cleveland Model is unique in that it secured commitments from the anchor institutions to purchase the products of the worker cooperatives. The first three cooperatives are the nation's most eco-efficient commercial laundry, a solar installation business, and an urban vegetable garden. The worker-owners are drawn from the impoverished communities surrounding the anchor institutions, and are typically previously unemployed residents. The cooperative structure spreads ownership to disenfranchised citizens and the institutional commitments mean that

these cooperatives—some quite significant in size—can be more successful than ordinary cooperatives. The model, which was pioneered by the Democracy Collaborative, has generated tremendous interest around the country, and many cities are now attempting to replicate it in one form or another.

This work is also representative of much other new economics activity in that it is taking place at a municipal, or city level. This has been the most innovative and vibrant scale for change since the downturn. Many cities' economies have been devastated, and local planners and politicians believe that conventional approaches such as enticing corporate investment with tax breaks are no longer feasible. They are therefore more willing to try new approaches than in the past. The hope of new economy theorists for municipal level activity is that a wide variety of services could be offered to maintain a sustainable local economy. These include municipal utilities and energy providers (many of which are already publically-owned), housing cooperatives, transportation systems, and regional food provisioning systems. Providing highly-efficient, low-cost public services to residents reduces their costs of living, makes possible a variety of low-carbon, high satisfaction lifestyles, and is a coping mechanism for a world in which traditional employment generation is structurally constrained (Rogers, 2011, see also Schor, 2010, for a similar account, although without the emphasis on municipally-provided services.)

There is also considerable activity in the area of finance. There are efforts to create publically financed state banks, like the Bank of North Dakota, whose mission is to finance enterprise within the state. There are also a number of local financing models that have emerged: residents of a town come together with small amounts of funding to keep a store operating, or to fund a new business. They are both customers and very small-scale owners ($1000 or less). The impact investment and Slow Money movements create long-term investment in sustainable agriculture or other areas, in which investors receive returns only over a long period. Crowdsourcing of finance, through a variety of platforms such as Kickstarter and Crowdfunder, is generating a rapidly growing pool of finance outside of established financial institutions. In 2013, an estimated 6 billion was generated through various types of crowdfunded schemes. Peer-to-peer lending is currently growing at more than 100% per year (Wikipedia, 2013c).

The online peer-to-peer sharing economy is also expanding rapidly (Schor and Fitzmaurice, 2014). Platforms include Couchsurfing and AirBnB, which are lodging sites; RelayRides, ZimRide, GoLoco and other car and ride sharing sites; time banking, Zaarly and TaskRabbit, which are service provisioning sites. Other platforms facilitate free, or "gift" exchange such as Freeycle and Yerdle. Share Some Sugar and NeighborGoods enable neighborhood sharing of durable consumer goods. There are also sites that organize the sharing of space or land for gardening, office work and storage. These sites are expanding rapidly. For example, in 2012 AirBnB reached a cumulative total of 10 million bookings since their founding. (https://www.airbnb.com/10-million?cdn_locale_redirect=1) Whether these platforms are in fact part of a new economy has become a topic of debate (Slee, 2013; Golumbia, 2013; Schor and Fitzmaurice, 2014). However, it seems that at least the non-profit platforms are aiming to reduce carbon footprints, build social ties and expand access to economic assets and services, which aligns their aims with those of other new economy innovations. (Actual impacts on carbon footprints remain unexplored, with a few exceptions. See Martin and Shaheen (2010). These sites also frequently employ a discourse of new economy.

5. Conclusions: the future of new economics

The "new economy" has not yet been quantified, although it is clear that it is growing, especially in the food and energy sectors. But it remains marginal, emergent and counter-cultural. However, its ultimate impact may be less in the exact models and forms that it pioneers than in its narrative of possibility of change. For it is only when there is enough belief that "there is an alternative," to subvert Thatcher's famous phrase, that enough momentum in the system can be harnessed to create the social movement and political power required to actually transform an economic system. In my view, the new economics movement is interesting because it has created an alternate frame for climate protection. But it is not a typical climate sustainability narrative, not only because it rejects tradeoff thinking but also because it does not prioritize climate or ecological impacts over human needs.

The foregoing analysis of the dominance of concerns about jobs and economic well-being post-2008 suggest that to be successful, new narratives or framings must address both economic and environmental challenges. We can only hope that one among the approaches to climate will be successful in this regard, so that out of the ashes of economic distress, a sustainable alternative can in fact emerge.

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